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Targeted v.s. Universal Fiscal Policy: A HANK Approach

1. Model and Methods

1.1 Model

The setup is a standard New Keynesian model with incomplete markets and unemployment risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms meet and match for labor decisions (Ravn and Sterk 2021), and the rest of the setup follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity, employment, and discount factor.

Households enjoy utility over consumption c_{it} and labor ℓ_{it}

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta_{it}^t \left(\frac{c_{it}^{1-\frac{1}{\gamma}} - 1}{1 - \frac{1}{\gamma}} - \psi \frac{\ell_{it}^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}} \right) \right],$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

A household faces unemployment risk $s \in \{E, U, V\}$, idiosyncratic labor productivity shocks e_{it} with $\mathbb{E}[e_{it}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, where $\beta_L < \beta_H$. The unemployment risk classifies households as employed or unemployed, with or without unemployment insurance.

Employed workers earn labor income $(1 - \tau_\ell)w_t e_{it} \ell_{it}$, where w_t is the wage rate and τ_ℓ is a fixed labor income tax rate. While, unemployed receive benefits b_t , that lasts on

average T_b periods, or a safety-net payment T_t . Let $y_t(s, e)$ be this non-financial income,

$$y_t(s, e) = \begin{cases} (1 - \tau_t)w_te, & \text{if employed;} \\ b_t, & \text{if unemployed;} \\ T_t, & \text{if needy.} \end{cases}$$

Households save into liquid assets a_{it} with real return $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$, where i_t represents the nominal interest rate and π_t denotes inflation. They also receive dividends d_{it} from firms. Their budget constraint at time t reads:

$$c_{it} + a_{it} = (1 + r_t)a_{it} + d_{it} + y_{it}(s, e).$$

Incomplete markets impose a borrowing constraint $a_{it} \geq \underline{a}$. Frictions in the labor market imply households delegate their labor supply decisions to labor unions, and thus take hours worked as given.

We solved the household problem using the Endogenous Grid Method (EGM) (Carroll 2006). The Bellman Equation for each individual i is:

$$\begin{aligned} V_t(s, \beta, e, a) = \max_{c, a'} & \left\{ \frac{c^{1-\frac{1}{\gamma}} - 1}{1 - \frac{1}{\gamma}} + \beta \mathbb{E}_t [V_{t+1}(s', \beta', e', a' | \cdot)] \right\} \\ \text{s.t.} \quad & c + a' = (1 + r_t)a + d_t + y_t(s, e) \\ & a' \in [\underline{a}, \infty), \end{aligned}$$

which means that the marginal value of the assets V_a is

$$V_a = \frac{\partial V_{t+1}}{\partial a'} = \frac{\partial V}{\partial c} \cdot \frac{\partial c}{\partial a} = (1 + r_{t+1})c^{-\frac{1}{\gamma}},$$

and the Envelope Condition tells us that

$$u'(c_t) = c_t^{-\frac{1}{\gamma}} = \beta \mathbb{E}_t [V_a^{t+1}] = \beta \mathbb{E}_t \left[(1 + r_{t+1})c_{t+1}^{-\frac{1}{\gamma}} \right].$$

Thus, we can create an inverse backwards iteration function of V_a^t with respect to V_a^{t+1} , which is the central idea of the EGM.

The unemployment risk is assigned by transition probabilities $\mathbb{P}(s' | s)$, where employed households may lose their job with probability λ , unemployed households may find employment with probability f , and lose the benefit with probability $1/T_b$, where T_b is

the average duration of the benefit.

$$\Pi_s = \mathbb{P}(s' | s) = \begin{bmatrix} 1 - \lambda & \lambda & 0 \\ f & 1 - f - 1/T_b & 1/T_b \\ f & 0 & 1 - f \end{bmatrix}.$$

Thus, the steady-state unemployment with and without the benefit is:

$$U_{ss} = \frac{f \cdot T_b}{1 + f \cdot T_b} \cdot \frac{\lambda}{\lambda + f}, \quad V_{ss} = \frac{1}{1 + f \cdot T_b} \cdot \frac{\lambda}{\lambda + f}.$$

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs

$$Y_t = \left(\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{jt} = Z_{jt}n_{jt}$. They hire labor at wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu W_t$, and the real wage equals $w_t = \frac{1}{\mu}$ at every date. Firms' dividends are taxed at the same rate τ_ℓ as labor income, so $d_t = (1 - \tau_\ell)(Y_t - w_t L_t) = (1 - \tau_\ell) \left(1 - \frac{1}{\mu}\right) Y_t$, where $L_t = \mathbb{E}[\ell_{it}]$ is the aggregate labor.

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to adjustment costs. We assume that unions allocate all labor hours uniformly across agents, so $\ell_{it} = \ell_t$ (Auclert, Rognlie, and Straub 2024).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $C_t = \int_0^1 c_{it} di$, discounted by the average discount factor $\bar{\beta}$, with Rotemberg (1982)'s adjustment costs. Thus results in a Phillips curve for wage inflation $\pi_t^w = (1 + \pi_t)w_t/w_{t-1}$ given by:

$$\pi_t^w = \kappa \left(\psi L_t^{\frac{1}{\phi}} - \frac{1 - \tau_\ell}{\mu C_t} \right) + \bar{\beta} \pi_{t+1}^w$$

Since prices are a constant markup over wages at all dates, price inflation is equal to wage inflation at all times.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_{t-1}^* + \phi\pi_{t-1}$, where r^* is the neutral real interest rate, and $\phi > 1$ (Taylor 1993). This assumption is very common in New Keynesian models and fits the data very well.

In the governmental sphere, we have a government that only collects taxes, makes transfers, and incurs debt; therefore, it does not purchase public goods, $G_t = 0$. The government's budget constraint is characterized by

$$b_t U_t + T_t V_t + (1 + r_{t-1})B_{t-1} = B_t + \tau_t w_t L_t,$$

which means that, in steady-state, $\tau w L - r B - b U - T V = 0$.

For dynamic simulations, we intend to apply two regimes: one financed by debt, with T fixed and $B_t = (1 + r)B + T + b(1 - L) - \tau w L$, and another financed by taxes, with B fixed and T variable. For now, we consider a fixed T_t and b_t and b_t , treated as exogenous, to calculate the IRFs based on it.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). We define $\Lambda_t(s, e, a)$ as the household distribution over a given period of time. First, the asset market reaches equilibrium when household savings equal government bonds

$$A_t = \int a_{i,t} d\Lambda_t(s, e, a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$L_t = \int \mathbb{1}\{s_{i,t} = E\} d\Lambda_t(s, e, a) = 1 - U_t - V_t;$$

and the goods market clear when $Y = C$ in the steady (because $\pi \approx 0$), due to the no government spending and no investment assumptions, but, due to the Rotemberg's adjustment cost, outside the steady-state it clears when:

$$Y_t = \left(\int_0^1 y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}} = \int c_{it} d\Lambda_t(s, e, a) = C_t.$$

1.2 Computational Methods

The main method is the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021), which can be used in a `Python` library. The central idea is to summarize the model by

a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous variables.

The method is built to find General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G} d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1} \mathbf{F}_{\mathbf{Z}} d\mathbf{Z}.$$

We are interested in calculating $dC = \mathbf{G}_{C,b} db$, the total effect of the change in unemployment insurance on household consumption. To do this, we divide the model into blocks so that the households are heterogeneous.

The main method for resolving heterogeneity is the Endogenous Grid Method (EGM) (Carroll 2006), which solves the value function iteration using the Euler equation, without solving the family maximization problem.

We assume a Rouwenhorst Markov Process for income $\log(y_t) = \rho_e \log(y_{t-1}) + \varepsilon_t^e$, with $\varepsilon_t^e \sim N(0, \sigma_e^2)$, which is more accurate than Taucher for highly persistent processes. We also assume a log-spaced grid for assets, in order to generate more robustness near the credit constraint.

We solve the steady-state equilibrium computationally by Brent's method, calibrating the parameters $(\sigma, f, s, \rho_e, \sigma_e, n_E, a_{min}, a_{max}, n_A, Z, \pi, r, \tau, b, B, \phi, r^*, \mu, \kappa)$, and estimating (β, w, Y) aiming at clearing the asset and goods markets and the NKPC. The labor market was left without defined criteria, as were U and the labor market, for the purposes of sanity checks, and everything went well.

2. Results

We solve the model at the calibrated steady state and compute both partial-equilibrium (PE) and general-equilibrium (GE) Jacobians with respect to the unemployment benefit b and to the transfers to needy T , treating it as the fiscal instrument.

Steady-state distribution. The stationary wealth distribution exhibits a sharp concentration near the borrowing constraint. A small number of households hold zero liquid assets and are therefore hand-to-mouth (HtM), while the Lorenz curve confirms, but less than what the 2019 Survey of Consumer Finances shows for the US economy. These two features, a non-trivial HtM share and a sizeable pool of unemployed households, are the key distributional facts that make fiscal targeting welfare-relevant in this economy.

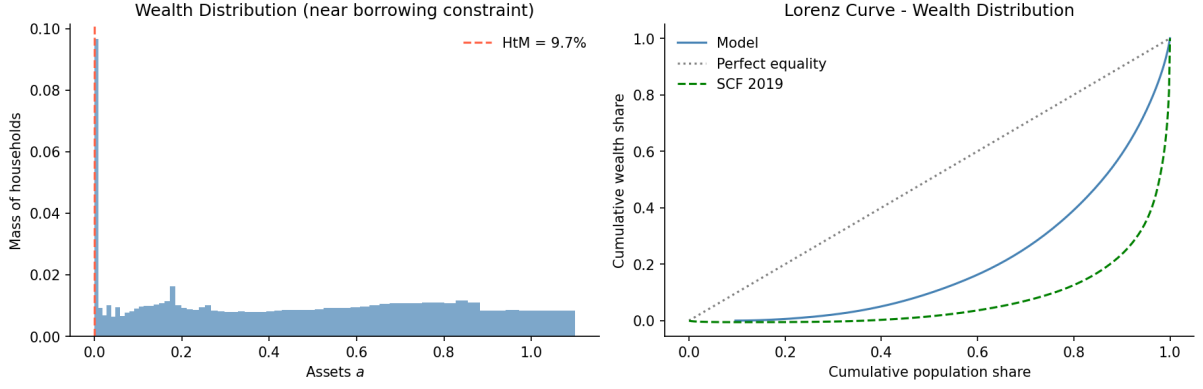


Figure 1: Wealth Distribution

Consumption policy function. The consumption policy function is nearly linear in assets for both employed and unemployed households at the median productivity level. The two schedules are nearly parallel, with the unemployed consuming slightly less at every asset level due to the lower benefit income $T < b < (1 - \tau)w\bar{e}$. The slope of both curves is close to the real interest rate r , indicating that high-asset households are close to permanent-income consumers and will save most of any windfall, a finding consistent with Bartal and Becard (2024) and central to the targeting argument.

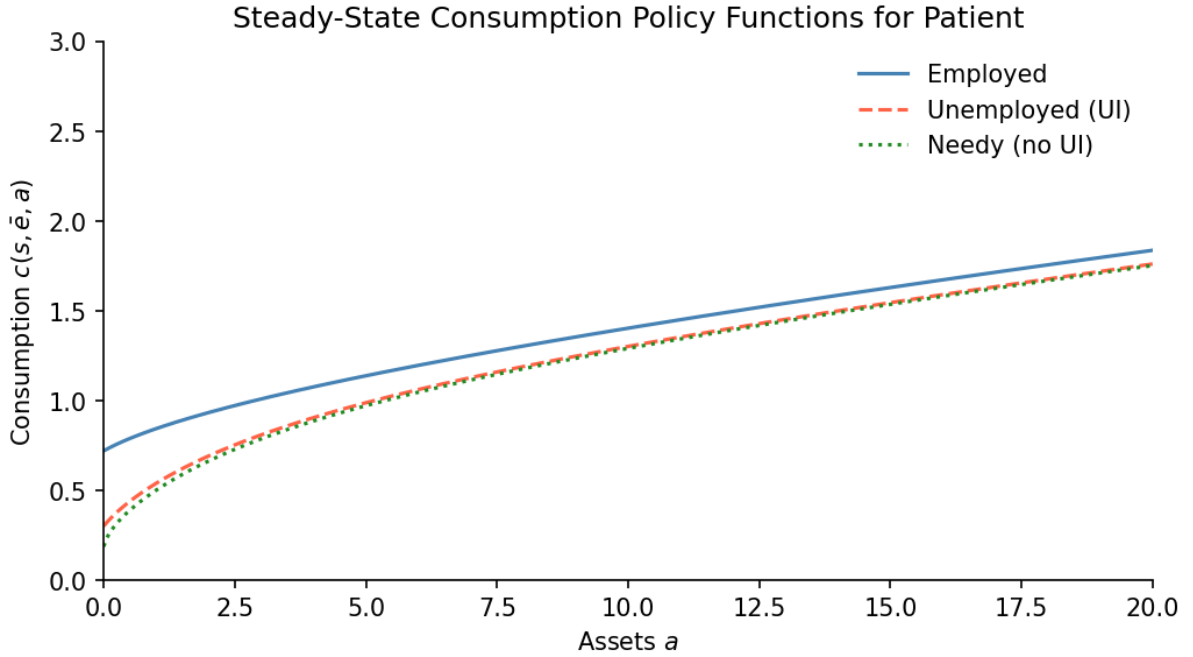


Figure 2: Steady-State Consumption Policy Function for Patient

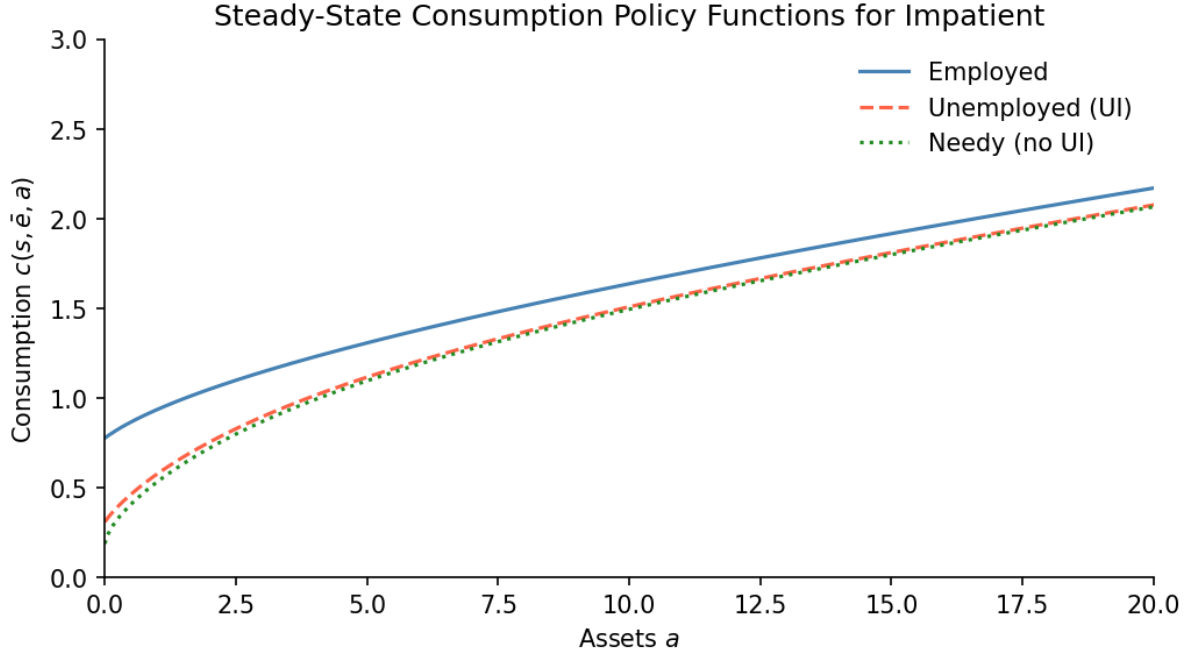


Figure 3: Steady-State Consumption Policy Function for Impatient

Intertemporal MPC profile. The PE intertemporal MPC of aggregate consumption with respect to b decays rapidly, while it takes longer for T to fall. This profile reflects the fact that unemployed households are disproportionately constrained and spend their additional income quickly. The monotone decay, without the hump shape typical of models with patient savers, is consistent with the one-asset structure: there is still no liquid-illiquid wedge that would delay consumption.

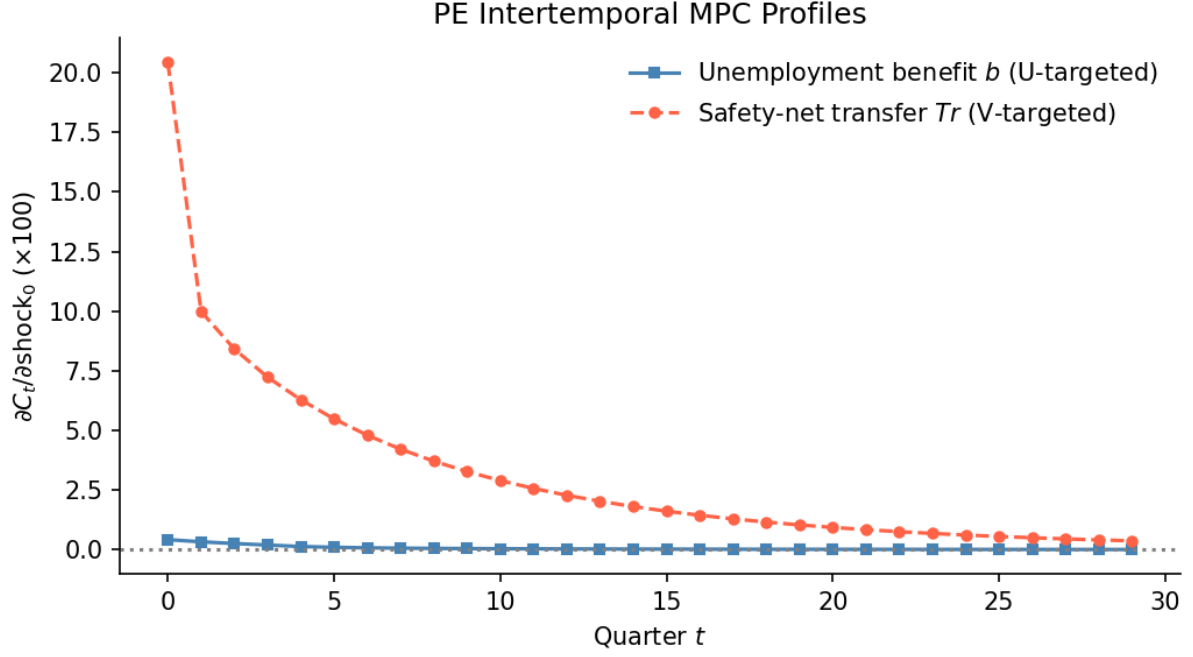


Figure 4: Partial Equilibrium MPC Profiles

General equilibrium impulse responses. We compute GE impulse responses to a persistent 1% rise in b (AR(1) with $\rho = 0.4$). The GE output and consumption multipliers coincide with the PE impact, indicating that GE amplification through wage and interest-rate channels is negligible for this shock at the current calibration. The unemployment rate IRF is essentially zero, since employment transitions are still exogenous in this model, and the GE wage response is too small to generate meaningful labor-market feedback.

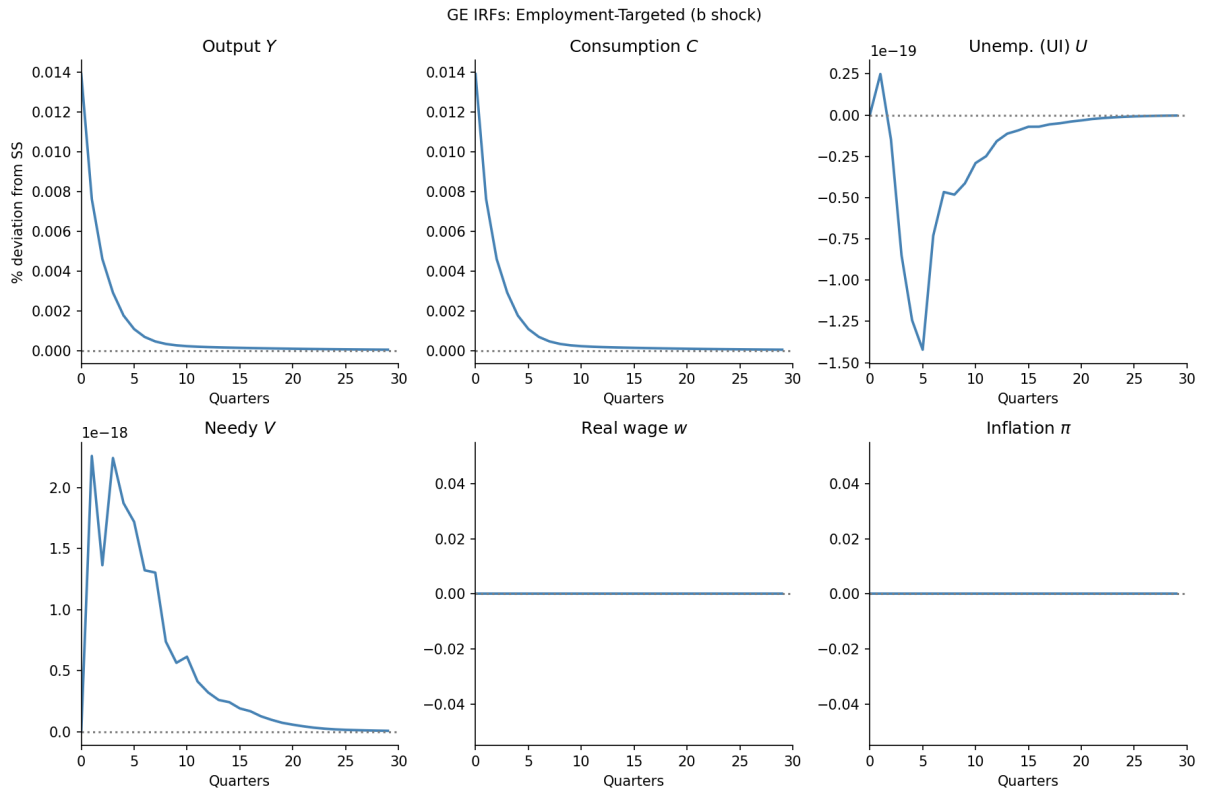


Figure 5: IRFs: Employment-Targeted (shock in b)

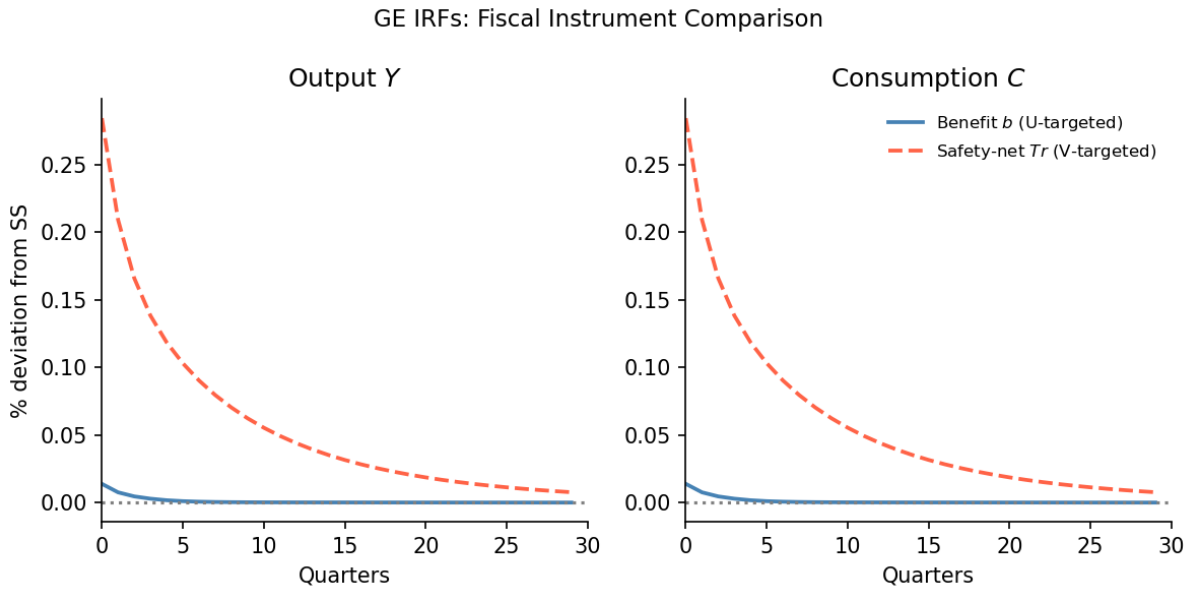


Figure 6: IRFs: Comparison (shock in b versus in T)

These results establish a baseline: an employment-targeted transfer achieves a well-identified direct income effect concentrated among high-MPC households, with limited GE amplification. The next step, comparing this to a cost-equivalent universal transfer,

will reveal whether targeting dominates on aggregate output grounds, or whether the broader reach of universal transfers compensates for their lower average MPC.

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Appendix: Calibration Table

Table 1: Calibration

Parameter	Description	Value	Source / Target
<i>Households</i>			
γ	Elasticity of Intertemporal Substitution	0.5	Standard in HANK literature
β_H	Discount factor for the Patient	0.9598	Estimated: asset market
β_L	Discount factor for the Impatient	0.9095	Difference set to 0.05
<i>Labor market</i>			
f	Job-finding probability (quarterly)	0.3	$U^{ss} + V^{ss} = s/(s + f) = 6.25\%$
s	Separation probability (quarterly)	0.02	$U^{ss} + V^{ss} = s/(s + f) = 6.25\%$
<i>Idiosyncratic income process</i>			
ρ_e	Persistence of log-productivity	0.966	Kaplan, Moll, and Violante (2018)
σ_e	Std. dev. of productivity innovation	0.5	Kaplan, Moll, and Violante (2018)
n_e	Number of Rouwenhorst nodes	7	Standard discretization
<i>Asset grid</i>			
$\underline{a}$	Borrowing constraint	0.0	No borrowing (McKay and Reis 2016)
$\bar{a}$	Maximum asset	200	Sufficient to cover top of dist.
n_a	Number of asset grid points	200	Accuracy near constraint
<i>Prices and aggregate</i>			
Y	Aggregate Output	1.0	Normalization
Z	Total Productivity	1.07	Estimated: labor market
π	Steady-state inflation	0.0	Standard SS assumption
w	Real wage	0.961	Estimated: NKPC ($w = Z/\mu$)
<i>Government</i>			
τ	Flat labor income tax rate	0.25	Kaplan, Moll, and Violante (2018)
b	Unemployment benefit (flat)	0.3	Shock variable of interest
T	Safety-net transfer to needy (flat)	0.1	Shock variable of interest
B	Government bond supply	1.2	$B/Y = 120\%$
<i>Monetary policy</i>			
r^*	Neutral real interest rate	0.5%	2% annual average (US data)
ϕ	Taylor rule inflation coefficient	1.5	Standard (Taylor 1993)
<i>Firms</i>			
μ	Price markup	1.11	10% markup (Basu and Fernald 1997)
κ	NKPC slope (Rotemberg)	0.1	Hazell et al. (2022)
κ_w	Wage-NKPC slope (Rotemberg)	0.1	Equal to κ

EIS = 0.5 is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).